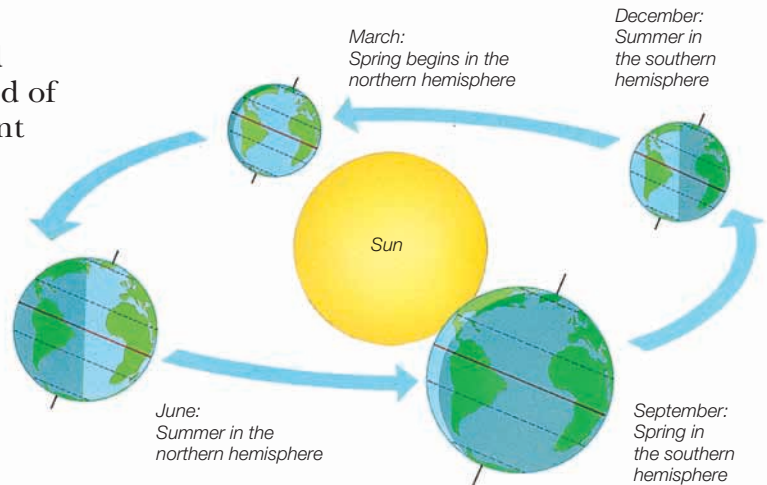


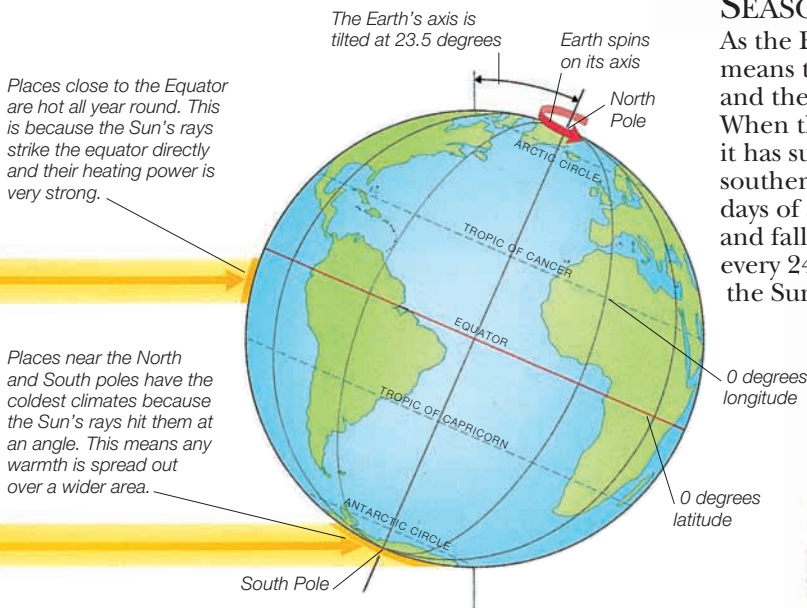
CLIMATE AND VEGETATION

CLIMATE IS THE AVERAGE PATTERN of weather and temperature in a particular area over a long period of time. Similar types of climate are found in different places around the world. For example, there are regions of hot, dry desert in Africa and North America, as well as across central Australia. It is a region's climate, together with its physical landscape, that determines the kind of vegetation, or plant life, that is usually found there. Cold areas near the poles and icy mountain peaks support little, or no, vegetation. Hot, wet rain forests near the equator, however, encourage the fast growth of a variety of plants.



SEASONS OF THE YEAR

As the Earth travels around the Sun, the tilt on its axis means that each place leans gradually nearer the Sun, and then farther away from it. This causes the seasons. When the northern hemisphere leans toward the Sun it has summer. When it tilts away it has winter. In the southern hemisphere this is reversed. Between the warm days of summer and the cold days of winter come spring and fall. The Earth also spins on its axis, turning once every 24 hours to give us day and night. The side facing the Sun has day, while the other side has night.



LATITUDE AND LONGITUDE

A region's climate is influenced by how far to the north or south of the equator it lies. This is called its latitude. The equator, an imaginary line running around the Earth, lies at 0 degrees latitude. Other lines of latitude include the Tropics of Cancer and Capricorn. Regions around the equator are the hottest in the world, while the closer to the poles you go, the colder it gets. There are also longitude lines that run from north to south, known as meridians.



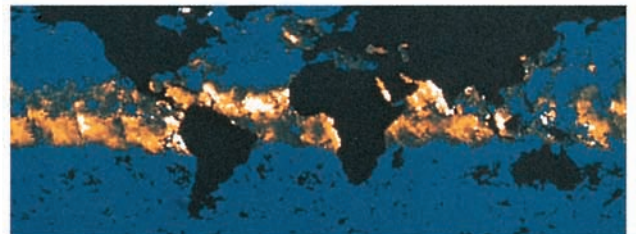
LAND AND SEA

The climate of a region is affected by altitude – how high a place is above sea level. The higher a place, the colder its climate, even if it lies near the equator or the Tropics, like these Atlas Mountains in Morocco. Another important influence is how close a place is to the sea. The sea warms and cools more slowly than land, so coastal areas often have fewer extremes in temperature.



RAINFALL

The amount of rainfall a place receives during the year greatly affects its vegetation as well as its climate. Plants need water to make their own food and will thrive in the warm, wet climate of a tropical rain forest, shown here in Costa Rica. Where rainfall is very low, in deserts and polar regions, only a few plants manage to survive. In other places, the amount of rainfall varies with the seasons.



CHANGES IN WORLD CLIMATE

The world's climate can be changed by both natural as well as human events. When Mt. Pinatubo, a volcano in the Philippines, erupted in 1991, it threw ash and dust high into the atmosphere. Locally, this caused dark skies, heavy rainfall, and high winds. The distance the ash was carried can be seen from this satellite photo. Equally, events such as the massive oil fires in Kuwait, started during the Gulf War, can have a damaging effect on climate.